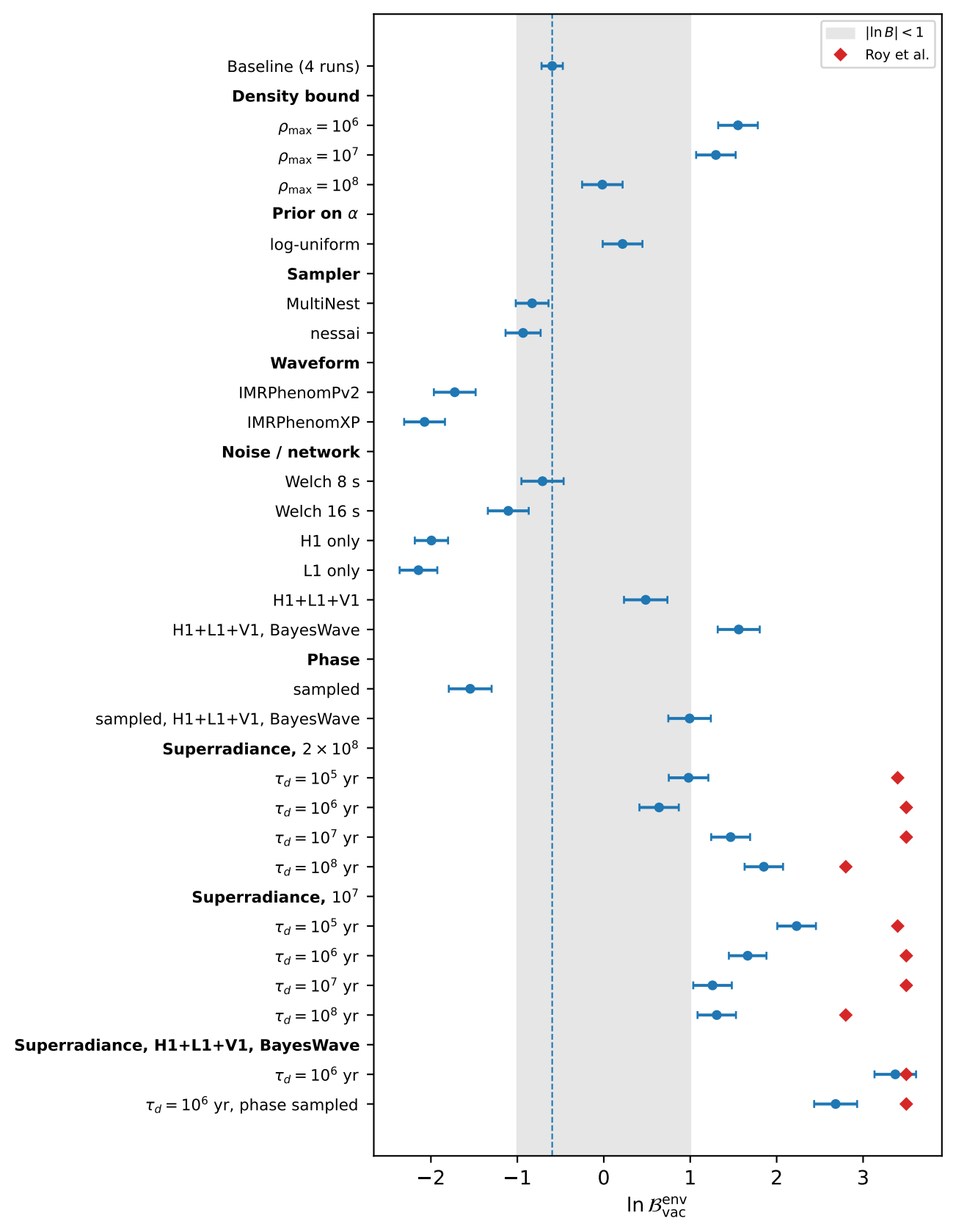
$|\ln B| < 1$
Roy et al.
Baseline (4 runs)
**Density bound**
$\rho_{\max} = 10^6$
$\rho_{\max} = 10^7$
$\rho_{\max} = 10^8$
**Prior on $\alpha$**
log-uniform
**Sampler**
MultiNest
nessai
**Waveform**
IMRPhenomPv2
IMRPhenomXP
**Noise / network**
Welch 8 s
Welch 16 s
H1 only
L1 only
H1+L1+V1
H1+L1+V1, BayesWave
**Phase**
sampled
sampled, H1+L1+V1, BayesWave
**Superradiance,** $2 \times 10^8$
$\tau_d = 10^5$ yr
$\tau_d = 10^6$ yr
$\tau_d = 10^7$ yr
$\tau_d = 10^8$ yr
**Superradiance,** $10^7$
$\tau_d = 10^5$ yr
$\tau_d = 10^6$ yr
$\tau_d = 10^7$ yr
$\tau_d = 10^8$ yr
**Superradiance, H1+L1+V1, BayesWave**
$\tau_d = 10^6$ yr
$\tau_d = 10^6$ yr, phase sampled
−2
−1
0
1
2
3
$\ln \mathcal{B}^{\text{env}}_{\text{vac}}$